

MAT 534 HW04

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Due 10/01

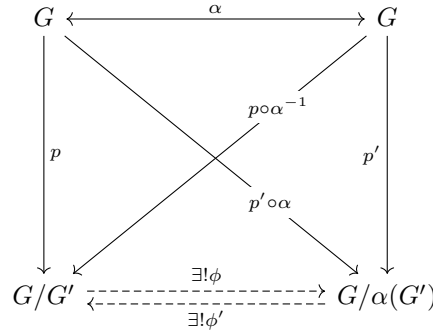
Problem 1. A **characteristic subgroup** of a group is a subgroup $H \leq G$ such that $\alpha(H) = H$ for every automorphism $\alpha \in \text{Aut}(G)$. Show that the center Z and the commutator subgroup G' are characteristic subgroups of G .

Solution. Let $\alpha \in \text{Aut}(G)$. Then if $a \in Z$, let $g \in G$ be arbitrary; we can compute

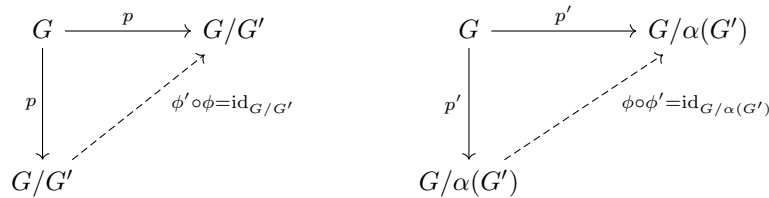
$$\alpha(a)\alpha(g) = \alpha(ag) = \alpha(ga) = \alpha(g)\alpha(a),$$

so $\alpha(a)$ commutes with $\alpha(g)$ for all $g \in G$. Since α is a bijection on G , every $h \in G$ can be represented as $\alpha(g)$ for some $g \in G$, so $\alpha(a)$ commutes with every element of G , meaning it is in Z ; that is, $\alpha(Z) \subseteq Z$. On the other hand, $\alpha^{-1} \in \text{Aut}(G)$ is also an automorphism, so $\alpha^{-1}(Z) \subseteq Z \implies Z \subseteq \alpha(Z)$, giving the reverse inclusion, hence $\alpha(Z) = Z$.

For the commutator subgroup, let $\alpha \in \text{Aut}(G)$ and let $p: G \rightarrow G/G'$ and $p': G \rightarrow G/\alpha(G')$ be the projection homomorphisms. Then, the kernel of $p' \circ \alpha$ is G' because the kernel of p' is $\alpha(G')$ and the kernel of $p \circ \alpha^{-1}$ is $\alpha(G')$ because the kernel of p is G' , so by the universal property of quotients, there exist unique $\phi: G/G' \rightarrow G/\alpha(G')$ and $\phi': G/\alpha(G') \rightarrow G/G'$ such that the following diagram commutes:



Then, noting that $p = \phi' \circ \phi \circ p = p$ and $p' = \phi \circ \phi' \circ p' = p'$, one can use the universal property of quotients again in the following diagrams to conclude that $\phi' \circ \phi = \text{id}_{G/G'}$ and $\phi \circ \phi' = \text{id}_{G/\alpha(G')}$.



Thus $G/G' \cong G/\alpha(G')$ is abelian; G' is the minimal normal subgroup among all normal subgroups for which the quotient is abelian, so $G' \leq \alpha(G')$. Applying the same to the automorphism α^{-1} gives $G' \subseteq \alpha^{-1}(G') \implies \alpha(G') \subseteq G'$, proving that $\alpha(G') = G'$. $\square$

Problem 2. Let $\phi: G \rightarrow \text{Aut}(G)$ be the map that associates to any $g \in G$ the corresponding inner automorphism $\phi(g) = \alpha_g$, where $\alpha_g(x) = gxg^{-1}$ for all $x \in G$.

- (a) Show that ϕ is a homomorphism.
- (b) Show that the kernel of ϕ is the center $Z \leq G$.
- (c) Let $\text{Inn}(G)$ be the image of ϕ . Show that $\text{Inn}(G)$ is a normal subgroup of $\text{Aut}(G)$.

Solution. (a) If $g, h \in G$ then $\phi(g)\phi(h) = \alpha_g \circ \alpha_h$, which is the map $x \mapsto g(hxh^{-1})g^{-1} = \alpha_{gh}(x) = \phi(gh)(x)$, so ϕ is a homomorphism.

(b) g is in the kernel of ϕ if $\phi(g)$ is the identity automorphism, i.e. $gxg^{-1} = x$ for all $x \in G$; this is exactly what it means for g to be in the center Z , so the kernel of ϕ is Z .

(c) Let $\alpha \in \text{Aut}(G)$ and $g \in G$. Then the conjugate $\alpha \circ \alpha_g \circ \alpha^{-1}$ is given by

$$(\alpha \circ \alpha_g \circ \alpha^{-1})(x) = \alpha(g\alpha^{-1}(x)g^{-1}) = \alpha(g)x\alpha(g^{-1}) = \alpha_{\alpha(g)}(x) \in \text{Inn}(G),$$

so $\text{Inn}(G)$ is closed under conjugation, hence is normal in $\text{Aut}(G)$. □

Problem 3. Let $n \geq 2$. Show that A_n is the only subgroup of S_n of index 2.

Solution. If $n = 2$ then there is obviously only one index 2 subgroup (the trivial subgroup).

Now let $N \leq S_n$ be a subgroup of index 2. Since 2 is the smallest prime it is certainly the smallest prime factor of S_n , so by HW02 #6(e), N is normal in S_n .

If $n = 4$ then there are five conjugacy classes:

- $\{(1)\}$, with one element;
- 2-cycles, of which there are $\binom{4}{2} = 6$;
- 3-cycles, of which there are $\frac{4 \cdot 3 \cdot 2}{3} = 8$;
- 4-cycles, of which there are $\frac{4 \cdot 3 \cdot 2 \cdot 1}{4} = 6$; and
- products of two disjoint 2-cycles, of which there are $\frac{1}{2} \binom{4}{2} \cdot \binom{2}{2} = 3$

(one can verify these are all the conjugacy classes by checking that the total number of elements enumerated is 24). Since N is the union of some of these conjugacy classes and contains the trivial conjugacy class, numerically we need $1 + \sum |C| = 12$ where the sum is over the nontrivial conjugacy classes C contained in N . If either of the 6s are included then the remaining conjugacy classes, which have sizes 3, 6, and 8, must contribute 5 more elements; this is clearly impossible. Therefore the only way to get a sum of 12 is $12 = 1 + 3 + 8$, which are precisely the conjugacy classes containing even permutations, i.e. $N = A_n$.

If $n = 3$ or $n \geq 5$, the simplicity of A_n for these n implies that $N \cap A_n \leq A_n$ is either the trivial group or A_n . In the former case, N must contain at least one 2-cycle, hence contains the entire conjugacy class of 2-cycles, so $N = S_n$, which is a contradiction. Thus only the latter case is possible, where $N = A_n$ because $|N| = |A_n|$. □

Problem 4. Let $n \geq 3$, and let $N \leq A_n$ be the subgroup generated by all 3-cycles.

- (a) Show that $N \trianglelefteq A_n$.
- (b) Show that $N = A_n$.

Solution. The main idea is that a product of two 2-cycles can be written as a product of 3-cycles. To check this, we have 3 cases based on the number of shared elements in the 2-cycles; here, $a, b, c, d \in \{1, \dots, n\}$ are distinct.

- (i) $(a\ b)(a\ b) = (1)$ is trivially a product of (zero) 3-cycles.
- (ii) $(a\ b)(b\ c) = (a\ b\ c)$.
- (iii) $(a\ b)(c\ d) = (a\ c\ b)(a\ c\ d)$.

For any $\sigma \in A_n$, write σ as the product of an even number of 2-cycles $\sigma_1, \dots, \sigma_{2k}$ and pair off the 2-cycles:

$$\sigma = (\sigma_1\sigma_2)(\sigma_3\sigma_4) \cdots (\sigma_{2k-1}\sigma_{2k}).$$

Each product $\sigma_{2j-1}\sigma_{2j}$ is a product of 3-cycles, so $\sigma \in N$, hence $A_n \leq N$. Conversely $N \leq A_n$ because 3-cycles are even, hence every permutation in N is even, so $N = A_n$ (which obviously means it's normal in A_n). $\square$

Problem 5. Let $n \geq 4$. Show that there exists an injective group homomorphism $S_n \rightarrow \text{Aut}(A_n)$. What happens for $n = 3$?

Solution. Let S_n act on A_n by conjugation; this of course induces a homomorphism $S_n \rightarrow \text{Aut}(A_n)$. To check that it is injective suppose that $\sigma, \tau \in S_n$ act on A_n in the same fashion, i.e. $\sigma x \sigma^{-1} = \tau x \tau^{-1}$ for all $x \in A_n$. This means $\tau^{-1}\sigma \circ x = x \circ \tau^{-1}\sigma$, so $\tau^{-1}\sigma$ is in the center of A_n . If $f = \tau^{-1}\sigma$ is not the identity, then there exist $a, b \in \{1, \dots, n\}$ such that $f(a) = b$ and $a \neq b$. Since $n \geq 4$ there exist $c, d \in \{1, \dots, n\}$ distinct from each other and from a and b , so $f(b\ c\ d)$ maps a to b while $(b\ c\ d)f$ maps a to c , f doesn't commute with $(b\ c\ d)$, contradiction. Therefore $\tau^{-1}\sigma = (1) \iff \tau = \sigma$, so the homomorphism $S_n \rightarrow \text{Aut}(A_n)$ given by the conjugation action of S_n on A_n is injective.

However, when $n = 3$, $A_n \cong \mathbb{Z}/3\mathbb{Z}$ is abelian, so its center is nontrivial, which is where this breaks down; in particular $\text{Aut}(A_n)$ consists of two automorphisms, the identity and the one swapping $(1\ 3\ 2)$ with $(1\ 2\ 3)$. $\square$

Problem 6. Suppose that G is a simple group of order 60.

- (a) Show that G has exactly six Sylow 5-subgroups.
- (b) Show that the action of G on its Sylow 5-subgroups by conjugation gives an injective homomorphism $\phi: G \rightarrow S_6$, and that the image of ϕ is a subgroup of A_6 of index 6.
- (c) Show that $G \cong A_5$.

Solution. (a) By the Sylow theorem, the number of Sylow 5-subgroups divides $60/5 = 12$ and is $1 \pmod{5}$; these conditions together imply that G either has 1 or 6 Sylow 5-subgroups. The former implies that the Sylow 5-subgroup is normal in G , contradicting simplicity, so G must have six Sylow 5-subgroups.

(b) The kernel of ϕ is normal in G , so it must be either $\{1\}$ or G because G is simple. However, the kernel cannot be G because the conjugation action of G on its Sylow subgroups is transitive, so the kernel is the trivial group. $\phi(G)$ is a subgroup of A_6 because if it weren't, then the kernel of the

sign homomorphism $\text{sgn} : \phi(G) \rightarrow \{\pm 1\}$ would be an index 2 normal subgroup of G , which contradicts simplicity; and of course, by just counting the number of elements, $(A_6 : \phi(G)) = 360/60 = 6$.

(c) $\phi(G)$ is the stabilizer of the coset $\phi(G)$ when A_6 acts on the cosets of $\phi(G)$ by left multiplication, which is enough to imply that $\phi(G)$ is isomorphic to A_5 . $\square$

Problem 7. In this problem, we construct an exotic automorphism of S_6 .

- (a) Show that S_5 has exactly 6 Sylow 5-subgroups.
- (b) Show that the action of S_5 on its Sylow 5-subgroups gives an injective homomorphism $\phi : S_5 \rightarrow S_6$, whose image H is a subgroup of S_6 of index 6.
- (c) Let $H_k \subseteq S_6$ be the stabilizer of $k \in \{1, \dots, 6\}$. Show that H is *not* one of the subgroups $H_1, \dots, H_6$.
- (d) Show that the action of S_6 on the cosets of H gives an automorphism $\alpha : S_6 \rightarrow S_6$.
- (e) Show that α is *not* an inner automorphism.

Solution. (a) S_5 has more than one subgroup of order 5, since $\langle (1\ 2\ 3\ 4\ 5) \rangle$ and $\langle (2\ 1\ 3\ 4\ 5) \rangle$ are distinct subgroups of order 5; thus the number n_5 of Sylow 5-subgroups is at least 1. On the other hand, $n_5 \cong 1 \pmod{5}$ and $n_5 \mid 24$ by the Sylow theorem, so $n_5 \in \{1, 6\}$ by listing out the divisors; thus n_5 must be 6.

(b) The only normal subgroups of S_5 are $\{(1)\}$, A_5 , and S_5 . The conjugation action is transitive, hence the image of S_5 has more than two elements (at least one permutation in $\phi(S_5)$ sends 1 to j for each $j = 2, \dots, 6$), so $(S_5 : \ker \phi) > 2$ implies $\ker \phi = \{1\}$, hence ϕ is injective. The index of $\phi(S_5)$ in S_6 is $|S_6| / |S_5| = 6$.

(c) H is not any of the H_k because no Sylow 5-subgroup of S_5 is fixed under conjugation by all elements of S_5 .

(d) By HW02 #6(c), the kernel of α is $\sigma \in \bigcap_{g \in S_6} gHg^{-1}$; this is the trivial subgroup of S_6 because it is normal in S_6 (whose only normal subgroups are $\{(1)\}$, A_6 , and S_6) and has at most $|H| < |A_6|$ elements. Thus α is injective, so it is a bijection because its domain and codomain have the same number of elements.

(e) Let the cosets of H be $H = C_1, \dots, C_6$, so the action of a permutation τ fixes H if and only if $\alpha(\tau) \in H_1$. If α were an inner automorphism, say conjugation by σ , then it would bijectively map $H_{\sigma^{-1}(1)}$ to H_1 ; however, α instead maps H to H_1 , and part (c) established that H can't be equal to $H_{\sigma^{-1}(1)}$ for any $\sigma \in S_6$, which proves that α is not an inner automorphism. $\square$

Problem 8. Let G be a solvable finite group. Show that there is a chain

$$G = N_0 \supseteq N_1 \supseteq \cdots \supseteq N_r = \{1\}$$

such that N_i/N_{i+1} is cyclic for every $0 \leq i \leq r-1$.

Solution. The following lemma will let us insert groups into the chain of normal subgroups with abelian quotients guaranteed by the solvability of G :

Lemma 8.1. Let A be a group and $B \trianglelefteq A$ a normal subgroup such that $\bar{A} = A/B$ is abelian. Then there exists a chain

$$A = N_0 \supseteq N_1 \supseteq \cdots \supseteq N_r = B$$

such that N_i/N_{i+1} is cyclic for every $0 \leq i \leq r-1$.

Proof. Since $\bar{A} = \bar{N}_0$ is abelian, it is isomorphic to $(\mathbb{Z}/p_1^{e_1}\mathbb{Z}) \times \cdots \times (\mathbb{Z}/p_r^{e_r}\mathbb{Z})$ for some primes p_i and nonnegative integers e_i . Then, let $\bar{N}_j$ be a subgroup of $\bar{N}_{j-1}$ isomorphic to $\prod_{i=1}^{r-j} (\mathbb{Z}/p_i^{e_i}\mathbb{Z})$, so that $N_j/N_{j+1} \cong \mathbb{Z}/p_{r-j}^{e_{r-j}}\mathbb{Z}$ is cyclic. By the fourth isomorphism theorem, we can lift the chain $\bar{N}_0 \supseteq \cdots \supseteq \bar{N}_r = \{B\}$ to a chain of subgroups of A in which N_{j+1} is normal in N_j , and $N_j/N_{j+1} \cong \bar{N}_j/\bar{N}_{j+1}$ is cyclic, which is what we want. $\square$

Now, given a chain $G = N_0 \supseteq N_1 \supseteq \cdots \supseteq N_r = \{1\}$ such that successive quotients are abelian, we can use the lemma to insert subgroups between each N_j and N_{j+1} so that successive quotients are cyclic to get a chain of normal subgroups with the desired property. $\square$

Problem 9. Let G be a solvable group.

- (a) Show that every subgroup of G is solvable.
- (b) Let $\phi: G \rightarrow H$ be a homomorphism. Show that $\phi(G)$ is solvable.

Solution. (a) Let $H \leq G$. Define $G_0 = G$ and $G_n = G'_{n-1}$ to be the commutator subgroup of G_{n-1} , and similarly $H_0 = H$, $H_n = H'_{n-1}$. Then, $H_0 \leq G_0$, and if $H_{n-1} \leq G_{n-1}$, then a commutator in H_{n-1} is also a commutator in G_{n-1} , so $H_n = H'_{n-1} \leq G'_{n-1} = G_n$, hence by induction $H_n \leq G_n$ in general. Since G is solvable there exists N such that G_N is the trivial subgroup, which in turn means H_N is the trivial subgroup, hence H is also solvable.

(b) It is easy to check that commutators in G map to commutators in $\phi(G)$, and each commutator in $\phi(G)$ is mapped to by a commutator in G ; thus, if the derived series of G eventually stabilizes at the trivial subgroup, so does the derived series of $\phi(G)$. $\square$

Problem 10. Let $N \trianglelefteq G$. Show that G is solvable if and only if both N and G/N are solvable.

Solution. If G is solvable then N and G/N are solvable by Problem 9, since G/N is the image of the projection homomorphism $G \rightarrow G/N$.

If N and G/N are solvable, then we can lift a chain of normal subgroups of G/N with successive abelian quotients up to G via the fourth isomorphism theorem to get a chain $G = N_0 \supseteq \cdots \supseteq N_r = N$, and then append a chain of normal subgroups $N = N_r \supseteq N_{r+1} \supseteq \cdots \supseteq N_{r'} = \{1\}$ to the end to get that G is solvable. $\square$